Welcome to SM00146!

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What is this course about

- Systems Biology / Omics Analysis Course
  - For non-computationally-oriented PhD Students/Researchers
  - Don’t worry if you can’t code, almost everyone is on the same boat! ☺
- **Focus:** How we can use Omics Data and Computational Analysis to answer biological questions
  - Genomics, transcriptomics, metabolomics, proteomics
  - Systems biology
- Combination of theory + hands-on + scientific discussions

You do not need to become a computer scientist.

You should become able to understand, question, and use omics analyses in your own research.

What will not be discussed

- Experimental Parts:
  - Library prep
  - RNA-extraction
  - Randomization of samples, etc etc

(You guys know about this much better than I do)

- I will also not discuss about the quantification of the data, most companies provide this now
  - Drop me an email if you need to do this and I can guide you.
- Software or algorithm developments

What should you be able to do by the end?

You can read the ILO, but here's what I expect

Understand basic workflows of major omics analyses

Recognize why QC, normalization, metadata and statistics matter

Connect omics approaches to your own PhD project

Interpret PCA, volcano plots, heatmaps, enrichment, networks

Evaluate omics analyses in scientific papers

Communicate more effectively with computational collaborators

Know what to ask when you see, generate, or receive omics data.

How the course is organized

Theory	Hands-on	Journal club	Your research
What data represent Key assumptions Common methods What can go wrong	Real datasets Guided notebooks Run, inspect, interpret Ask why each step matters	Read papers critically Discuss omics design Opponent questions Constructive critique	Which omics layer helps? What question can it answer? What validation is needed?

The lecture is the “why”.
The notebook is the “how”.
The discussion is where you learn to judge the evidence.

Schedule and Important Dates

Monday	Tuesday	Wednesday	Thursday	Friday
Transcriptomics Lecture **9.15-12.00**	Proteo/Metabolomics Lecture **09.15-11.30**	Genomics Lecture **9.15-12.00**	Metabolomics+ MultiOmics Workshop **9.15-12.00**	Group Presentation and closing **09.15-12.00**
Transcriptomics Workshop **13.15-15.00**	Guest Lectures **13.15-15.00**	Genomics Workshop **13.15-15.00**	Group Discussion **13.15-15.00**	

- **18 September, 23.59: Deadline for written assignment**
- On Monday and Wednesday afternoon, I will stay until 16.00 in case you have any practical/workshop questions
- **Group discussion:** optional to be in class, but I will be available to answer questions

What this course is really about

Biological question → Experimental design → Omics measurement → Data processing → Statistical analysis → Biological interpretation → Validation

- A sophisticated analysis cannot rescue a weak biological question.
- Significant does not automatically mean biologically important.
- Omics produces evidence and hypotheses, not automatic truth.
- Interpretation and validation matter as much as computation.

Tools and resources

Course website / GitHub
SysBioPhD.multiomics.se

Posit Cloud
Jupyter notebooks + shared workspace

Canvas
Announcements + assignments + papers

- Use the course website as the main reference for notebooks and materials.
- Keep Posit Cloud invitation links private
- Download or save important work from notebooks after hands-on sessions.
- Course papers and instructions will be collected in Canvas

Hands-on sessions: what we expect

Do not just Execute

What am I doing?

Why does it matter?

How do I interpret it?

- No advanced coding experience is expected.
- Code is provided; focus on the purpose and interpretation.
- Work together, but make sure you understand your own results.
- Errors are part of computational work, not evidence that you are bad at it.

Learn the lingo!

- Knowing the lingo is important
  - Counts, TPM, Metadata, Design, PCA, Contrasts, FDR, Enrichment etc etc

- Knowing them will make your life easier (i.e. easier to google or use AI on something :D)

Use of AI tools

Helpful friend, but not a scientific authority

Good uses

- Explain code
- Understand errors
- Clarify concepts

Be careful

- Do not blindly trust code
- Check outputs
- Verify interpretation

Your responsibility

- Scientific reasoning
- Data interpretation
- Assignment conclusions

AI can help you write code. It cannot decide whether your biological conclusion is valid.

Course elements and assessment

Individual final assignment	Group journal club	Hands-on
One-page reflection: how and why omics / systems biology could support your research	20 min presentation + 10 min Q&A + opponent role	No submission needed, but please do them ☺

For the journal club, do not simply summarize the paper, evaluate it.

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Opening Question

- 2 mins, discuss it with your neighbor

You have unlimited money and can measure ONE omics layer in your project. Which one would you choose, and what biological question would it answer?

- **Foundations of omics:** understanding the major omics types, how data are generated from laboratory experiments through analysis, and the strengths and limitations of each approach.
  - Yes! The lectures!
- **Practical data analysis:** gaining hands-on skills for working with sequencing, transcriptomics, metabolomics, ~~metagenomics~~ and other omics datasets, including the use of databases and software.
  - Yes! The hands-on
- **Multi-omics integration:** learning how to combine transcriptomics, proteomics, genomics, metabolomics and ~~microbiome~~ data to obtain a more complete biological picture.
  - Yes! Last day will be about metabolomics and combining metabolomics and transcriptomics. Some of the journal club papers too.
- **Pathway and network interpretation:** strong interest in moving beyond lists of genes or metabolites toward biologically meaningful pathways, networks and systems-level interpretation.
  - Yes! Enrichment analysis

- **Experimental design and data quality:** interest in practical guidance on study design, batch effects, data quality and the interpretation of complex datasets.
  - Partially, data QC, batch effect. For study design, not easy to accommodate since the focus of the research is quite broad. We can speak separately
- ***Application to research:*** *several responses emphasized applying systems biology to specific research areas such as host-virus interactions, viral immunology, gut-body interactions,* ***drug discovery and development****.*
  - Same as above, practical application is not easy to incorporate, but we can discuss.
  - Guest lecture is from pharma, so drug discovery and development can be discussed
- **Emerging tools:** there was also interest in the use of databases and AI in systems biology and how computational approaches can support biological discovery.
  - We will use databases, but AI is pretty broad term :D
  - You can also ask the guest lecturer about this!

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